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Artificial Feeding

In subject area: Nursing and Health Professions

Artificial feeding is defined as the provision of nutrition through artificial means, either parenterally via total parenteral nutrition or enterally through tube feedings, such as nasogastric, gastrostomy, or jejunostomy tubes, for patients unable to consume food orally due to various medical conditions.

AI generated definition based on: Palliative Care (Second Edition), 2011

How useful is this definition? ☆ ☆ ☆

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Chapters and Articles

You might find these chapters and articles relevant to this topic.

Chapter

Withholding and Withdrawing Life-Sustaining Therapies

2011, Palliative Care (Second Edition)

Rohtesh S. Mehta, ... Robert M. Arnold

Artificial Nutrition

Artificial nutrition can be provided parenterally as total parenteral nutrition or enterally as tube feedings via nasogastric, gastrostomy, and jejunostomy tubes. More than 160,000 percutaneous endoscopic gastrostomy (PEG) tubes are placed annually in the U.S. Medicare patient population alone, and more than one third of nursing home residents with advance cognitive impairment receive artificial nutrition support via tube feeds.[11] The question whether to provide long-term artificial nutrition frequently arises in patients who are unable to take oral nutrition or who have anorexia or weight loss. Inability to take food orally is commonly the result of advanced catastrophic neurologic injury (e.g., stroke, ischemic encephalopathy,

diseases.

Over the past decade, a clinical consensus has developed that artificial nutrition should not be routinely provided to patients with advanced dementia because it does not provide clinical benefit. An evidence-based review of tube feeding in patients with advanced dementia did not find any survival benefit with a feeding tube. Rather, operative and perioperative mortality associated with feeding tube placement was substantial and ranged up to 25%.[12] Median survival after percutaneous endoscopic gastrostomy tube placement is less than 1 year, and mortality is 60% at 1 year and 80% at 3 years. Moreover, tube feeding does not decrease aspiration pneumonia, skin breakdown, or infections or increase functional status; rather, it can add to morbidity by increasing the risk of local and systemic infections.[12] It is not possible to know with certainty whether patients with advance dementia experience hunger, but their global decline in mental status suggests that they do not. The tube itself may be a source of discomfort.

Most patients with metastatic cancer and other terminal illnesses, in whom weight loss and decline in functional status are caused by the underlying disease, also do not benefit from artificial nutrition. On

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Chapter

Ethical Issues for Children With Cancer

2017, Ethical Challenges in Oncology

Kevin Madden, ... Patrick Zweidler-McKay

Withholding Nutrition and Hydration

Forgoing or discontinuing artificial nutrition and hydration (ANH) is supported by many medical professional associations including the American Medical Association [73] and AAP [74], but it is not without strong, differing opinions [75,76]. It is paramount that pediatric providers be aware of the deeply rooted, universal, and sacred importance of providing food and liquids to seriously ill children. For parents, providing nutrition—and its perceived benefits—to their child is one of the last things that is within their control; as

The ethical dilemma of continuing or discontinuing ANH can often be resolved by straightforward, open, and clear communication between the health-care providers and families. When explained that ANH can cause suffering in the form of fluid overload, electrolyte abnormalities, dyspnea, or clearly prolonging the dying process [77], families often understand and acknowledge that they do not want their child to suffer unnecessarily and will agree with

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Chapter

Palliative Care in Long-Term Care Settings

2011, Palliative Care (Second Edition)

Jennifer M. Kapo, Seema Modi

Evidence Summary: Risks and Benefits

Most literature about artificial nutrition and hydration in the long-term care setting focuses on patients with advanced cognitive impairment. Although there are no randomized, controlled studies that compare tube feeding with hand feeding, little evidence indicates that artificial nutrition and hydration prolong life, improve nutritional status, prevent aspiration, decrease development or promote healing of pressure ulcers, or improve a patient's comfort.[17] A large retrospective study showed that nursing home residents with dysphagia and advanced dementia did not have any survival benefit from receiving artificial nutrition and hydration, even after correcting for medical comorbidities. Another study reported that the median survival among elderly patients after percutaneous gastrostomy tube placement was 5.7 months, with 1-month mortality of 24% and 1-year mortality of 63%.[18]

A prospective cohort study of relatively younger residents (mostly without dementia) and observational studies of nursing home residents with advanced dementia found that artificial nutrition did not provide consistent improvement in weight or serum albumin. There is no evidence from existing studies that artificial nutrition and hydration via gastrostomy tube prevents aspiration.[17] Cohort

In addition to this lack of evidence of improved outcome, artificial nutrition and hydration have adverse effects at the end of life. There

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Chapter

Withholding and Withdrawing Life-Sustaining Therapies

2007, Palliative Care

Wendy G. Anderson, ... Robert M. Arnold

Artificial Nutrition

Artificial nutrition can be provided parenterally as total parenteral nutrition or enterally as tube feedings via nasogastric, gastrostomy, and jejunostomy tubes. More than 160,000 percutaneous endoscopic gastrostomy tubes are placed annually in the US Medicare population alone. The question whether to provide long-term artificial nutrition frequently arises in patients who are unable to take oral nutrition or who have anorexia or weight loss. Inability to take food orally is commonly the result of advanced dementia or catastrophic neurological injury (stroke, ischemic encephalopathy, traumatic brain injury). Decreased appetite and weight loss are also associated with cancer and other chronic diseases.

Over the past decade, a clinical consensus has developed that artificial nutrition should not be routinely provided to patients with advanced dementia, because it is not of clinical benefit. Although the consensus is not as clear, the same is probably true for patients with metastatic cancer and other terminal illnesses in whom weight loss and decline in functional status are caused by the underlying disease, not by lack of nutrition. Enteral nutrition does prolong life in certain neurological illnesses such as persistent vegetative state. In these cases, the decision to withhold or withdraw therapy should be based on the patient's definition of quality of life.

Finucane and colleagues performed an evidence-based review of tube feeding in patients with advanced dementia and found that the expected life span of these patients is short (13). Median survival is less than 1 year, and mortality is 60% at 1 year and 80% at 3 years. Observational studies have shown that tube feeding does not prolong life in advanced dementia, nor has it been shown to

tube feeding may increase aspiration. Patient comfort is another

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Chapter

Palliative Care in Long-Term Care Settings

2007, Palliative Care

Seema Modi, ... David Casarett

Ethical Issues Regarding Artificial Nutrition and Hydration

Ethical issues surrounding artificial nutrition and hydration often involve decision-making conflicts between the health care provider and patient's surrogates about the decision to initiate and the eventual decision to withdraw artificial nutrition and hydration. When a resident's wishes about artificial nutrition and hydration are known, it is reasonable to respect them, unless doing so will bring about obvious harm, such as increased respiratory distress or pain. Unfortunately, a resident's advance directive regarding wishes for artificial nutrition and hydration is often not available. When a patient's preferences are unknown, the surrogate should make a decision based on substituted judgment or knowledge of the patient's probable wishes. When the patient's probable wishes are not known, the surrogate must consider how a reasonable person would make a decision under the same circumstances. When artificial nutrition is being considered, clinicians should prepare patients and surrogates to consider a trial of therapy with eventual withdrawal if it does not achieve the desired effects.

Education of a resident's surrogate decision makers can greatly aid the process of decision making, but this education is often limited or nonexistent. Routine education at the time of nursing home admission may be helpful. Proactive education of decision makers

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Chapter

When Should Enteral Feeding by Percutaneous Tube Be Used in Patients With Cancer and in Patients With Non–Cancer-Related Conditions?

2013, Evidence-Based Practice in Palliative Medicine

Thomas Reid, Steve Pantilat

Legal and Ethical Issues

Law

Although the Supreme Court has ruled that artificial nutrition and hydration is a medical treatment that patients and surrogates may refuse, the evidence a surrogate must provide regarding a patient's preference varies among states.[45] In 20 states and the District of Colombia, where the evidentiary threshold for discontinuing tube feeds is higher than for other interventions, it is especially important to ask patients to document their preferences in advance and surrogates to thoroughly consider patients' goals before artificial nutrition and hydration is initiated.[46]

Ethics

A strong consensus supports withholding or withdrawing any life-sustaining intervention in the service of a patient's wishes. Although no ethical distinction exists between the two practices, clinicians and surrogate decision-makers may find it more difficult to withdraw an intervention such as enteral feeding than to withhold it in the first place. In most cases the argument for withdrawing artificial nutrition is actually stronger than that for withholding it, because experience has usually provided evidence of its specific failure to achieve the patient's goals.[47]

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Chapter

2013, Handbook of Clinical Neurology
Marie-Aurélie Bruno, ... Athena Demertzi

Guidelines

From a bioethical standpoint, withdrawing artificial nutrition and hydration is comparable to withdrawing mechanic ventilation, even if emotionally and psychologically these two actions may be perceived differently (Laureys, 2005). Despite the controversy as to whether artificial nutrition and hydration constitutes medical treatment (Bernat and Beresford, 2006), most of the medical community agrees that it is a medical therapy which can be refused by patients and lawful surrogate decision-makers (Steinbrook and Lo, 1988). Similarly, a national survey of Italian physicians' attitudes towards end-of-life decisions showed that 61% of the doctors questioned considered tube feeding to be a medical therapy and 66% believed that withdrawal of assisted nutrition and hydration might be appropriate depending on the patient's wishes (Solarino et al., 2011).

Such decisions in unconscious patients are only reasonable when the medical situation is considered as permanent and irreversible (Royal College of Physicians, 2003). In 1994, the Multi-Society Task Force on PVS defined the temporal criteria for irreversibility and currently states that if no recovery is observed within 3 months after a non-traumatic or 12 months after a traumatic accident, the condition of the patient can be denoted as permanent (The Multi-Society Task Force on PVS, 1994b).

For these problematic cases, relating to the value of life for patients who are conscious but not terminally ill, there are few ethical and legal guidelines about end-of-life treatment. Therefore, it is difficult to establish with any degree of certainty at what point after a brain injury this condition becomes irreversible and more longitudinal data are needed to confirm the criteria for, and timing of, realistic prognosis for recovery.

For example, the case of Wendland, a chronic traumatic patient in MCS with no advance directive, raised the ethical question on end of life in non-communicative patients. The court was asked to rule on his level of treatment. His wife claimed that, during informal conversations about end-of-life care with her and their children, her husband said that he never wished to live in such a state and that he would want his gastrostomy feeding tube removed to allow him to die. By contrast, Wendland's mother wished to maintain the gastrostomy tube feedings to keep him alive, claiming that he would wish to live under these circumstances. California's highest court

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Chapter

End of Life Support

2019, Encyclopedia of Cancer (Third Edition)

Kavitha J. Ramchandran, Jamie Von Roenn

Hydration and Nutrition at the End of Life

Several studies have examined the role of artificial nutrition at the end of life. A Cochrane meta-analysis concluded that there is insufficient evidence to recommend artificial nutrition at the end of life. Additionally, enteral feeding requires invasive procedures with risk of infection, bleeding, and pain, as well as complications of feeding such as diarrhea, aspiration, and electrolyte abnormalities. Parenteral feeding also has unique risks including infection, glucose control abnormalities, hepatic dysfunction, volume overload, and cholecystitis. With these risks in mind, artificial nutrition is not recommended at the end of life.

Artificial hydration is more controversial than feeding; some proponents believe that it may help with delirium at the end of life. However, there are recognized risks from hydration at the end of life, including volume overload, ascites, and pulmonary edema. If hydration is to be prescribed, low volumes are recommended with the goal of relieving symptoms, rather than delivering a set amount of fluid. Continuous hydration is generally not recommended.

At the end of life, patients often do not feel hungry, so it is appropriate to provide food and fluid based on the patient's desires. This is often called "pleasure feeding" and may involve small quantities offered at frequent intervals. If patients are having difficulty with swallowing, patients still benefit from frequent mouth care (keeping the oral cavity moist and clean) since xerostomia is a troubling symptom for patients with advanced illness. Discontinuation of oral intake by patients is a major source of emotional distress for families. Families may feel as if they are starving their loved one. It is important to reassure families that loss

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Nutritional treatment in the critically-ill complicated patient

2021, Nutrition and Bariatric Surgery

María Asunción Acosta Mérida, ... Gema M. Hernanz Rodríguez

Hyperglycemia

Hyperglycemia is a common complication of artificial nutrition due to the intravenous infusion of glucose and the metabolic stress associated with acute illness. The fat excess of obese subjects also contributes to insulin resistance, leading to hyperglycemia in a high proportion of these patients. The transcendence of hyperglycemia and plasma glucose variability is their relation to adverse clinical outcomes, including mortality [36]. Blood glucose measuring is recommended when the patient enters the ICU or when artificial nutrition is started and a minimum of every 4 h during the first 2 days of admission [3]. More frequent monitoring should be granted to unstable patients and, ideally, with arterial or venous samples. Capillary glycemic control is less reliable in the critical setting. Fig. 8.1 shows the proposed scheme for the management of nutritional support in critical obese patients.

Although the exact range of blood glucose levels is not definitely established, there is a general consensus that the goal is keeping values between 140 and 180 mg/dL, and the upper desired level should not exceed 180 mg/dL [3,35]. The main concern is avoiding hypoglycemia because although some trials offered promising results with lower blood glucose levels, especially in surgical patients (80–110 mg/dL) [37,38], these results were not confirmed in later publications. No diminished mortality could be demonstrated in the VISEP Trial despite the risk of severe hypoglycemia [39]. In fact, in the large NICE-Sugar Trial, where more than 6000 patients were included with a blood glucose target of 80–100 mg/dL, a tight glucose control group had higher mortality at 90 days (27.5% vs. 24.9%; P=.02) [40].

Treatment of choice for hyperglycemia in ICU is intravenous insulin, which must be adjusted to maintain blood glucose in the mentioned interval while reducing the risk of hypoglycemia. The amount of calories infused may influence insulin requirements, reducing both length (3.2±2.7 vs. 8±0.5 days) and dose (36.1±47.1 UI vs. 61.1±61 IU) [20]. A widely extended method for controlling hyperglycemia is starting intravenous insulin infusion at a minimum of 0.5 UI/h and

nutrition, to provide both glucose and insulin simultaneously [41].

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Chapter

History of surgery, metabolism, and nutrition therapy

2019, The Practical Handbook of Perioperative Metabolic and Nutritional Care

M. Isabel T.D Correia

The history of metabolism and nutrition

The first reports on the importance of artificial nutrition may date back to the ancient Egyptians and Greeks (3,500 years ago) who infused nutrients into the rectum to treat intestinal disorders, such as diarrhea. The solutions were mostly made from wine, milk, whey, and wheat broths, but sometimes they also contained eggs and brandy. President Garfield, from the United States of America who was gunshot in 1881, died of sepsis and cachexia after 80 days of painful suffering and unsuccessful attempts of rectal feeding [23]. For thousands of years, rectal feeding remained the method of choice, not necessarily effective, despite the efforts of feeding into the upper gastrointestinal tract reported since the 12th century. However, it was only in the 18th century that John Hunter used a hollow catheter and a syringe to infuse blended food (jelly, sugar, milk, wine, and water) into the stomach of a patient. Nonetheless, the importance of nutrition in man's life has been part of the history of Medicine with Hippocrates himself and his followers discussing how nutrition was related to health and disease. Although the famous widespread quote attributed to him "Let food be thy Medicine and Medicine be thy food" has been erroneously interpreted by westerners, it seems clear that food was valued by the Hippocratic Medicine [24].

The association between nutritional status and surgical outcomes

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Related terms:

End of Life Care, Advance Directive, Life Sustaining Treatment, Advance Care Planning, Symptom, Sedation, Enteric Feeding, Nursing Home, Palliative Therapy.

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